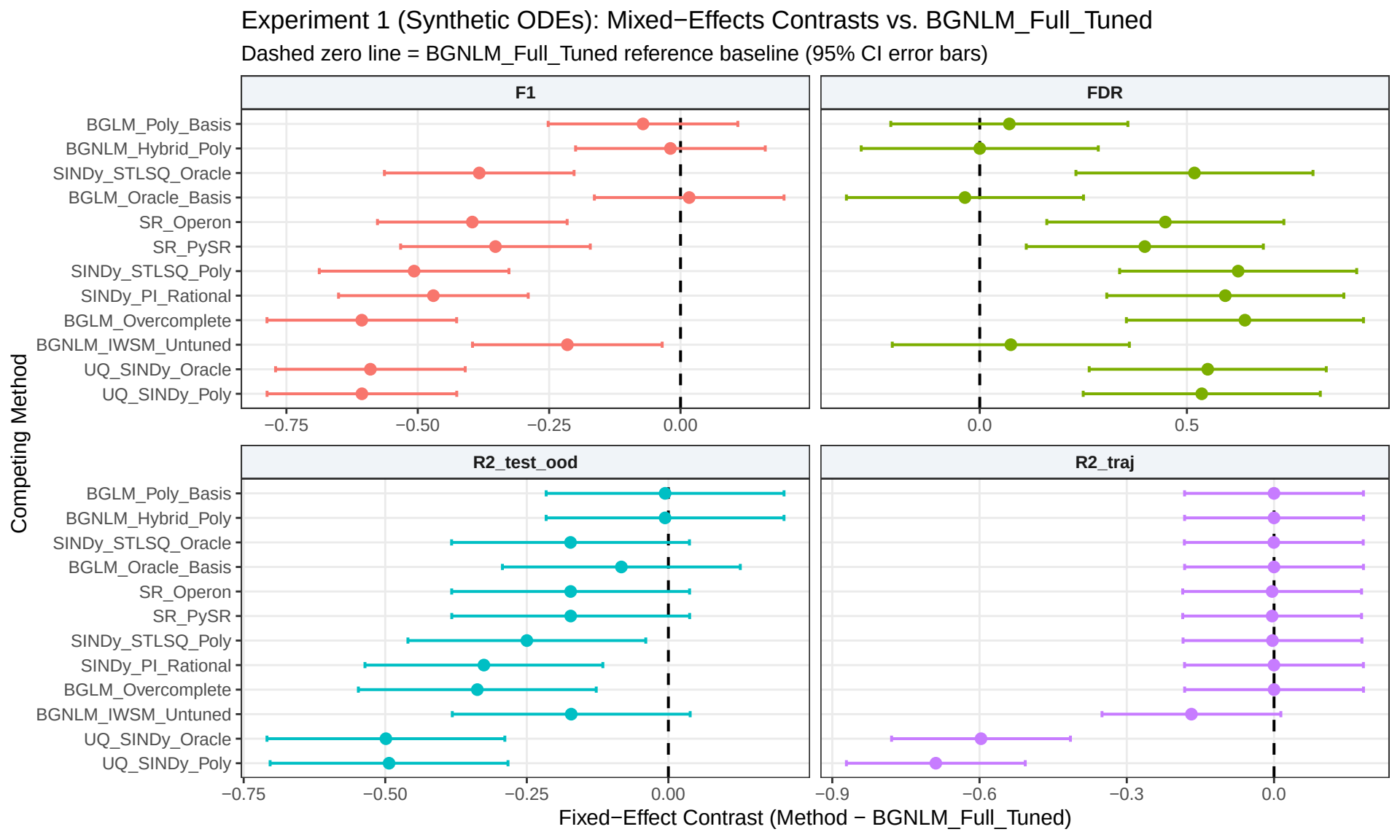
Experiment 1 (Synthetic ODEs): Mixed−Effects Contrasts vs. BGNLM_Full_Tuned
Dashed zero line = BGNLM_Full_Tuned reference baseline (95% CI error bars)
F1
FDR
R2_test_ood
R2_traj
BGLM_Poly_Basis
BGNLM_Hybrid_Poly
SINDy_STLSQ_Oracle
BGLM_Oracle_Basis
SR_Operon
SR_PySR
SINDy_STLSQ_Poly
SINDy_PI_Rational
BGLM_Overcomplete
BGNLM_IWSM_Untuned
UQ_SINDy_Oracle
UQ_SINDy_Poly
Competing Method
−0.75
−0.50
−0.25
0.00
0.0
0.5
−0.9
−0.6
−0.3
Fixed−Effect Contrast (Method − BGNLM_Full_Tuned)